F495 — honest refinement of the CP forcing: I WALK BACK the specific $\mathbb{Z}_3$ cube-root (equilateral-triangle) geometry of F493/F494 (the 3 generations = the 3 KW strata are a NESTED FILTRATION {dim $0 \subset 2 \subset 5$}, a LINEAR order, NOT a cyclic $\mathbb{Z}_3$ orbit — so 'generations at the 3 color cube-root corners' was over-specified, not forced), and I STRENGTHEN to the defensible core, which is a genuine forcing: CP violation is FORCED (generic, not fine-tuned) by TWO substrate facts — (A) the substrate is 2D-COMPLEX (Casey's sequential 2D-complex commits; D_IV^5 is a holomorphic bounded symmetric domain), which gives the irremovable PHASE (a REAL substrate gives real overlaps → real CKM → NO CP, verified), and (B) there are 3 generations (= N_c, ≥ the Kobayashi-Maskawa minimum), which is the minimum for the phase to be rephasing-irremovable (2 generations → one relative phase, removable → no physical CP; 3 → the rephasing-invariant triple product $\mathrm{Im}(\langle 1|2\rangle\langle 2|3\rangle\langle 3|1\rangle) \neq 0$). So CP violation is the OBSERVABLE SIGNATURE that the substrate is genuinely COMPLEX, not real — a real substrate would be a world with no CP and no matter/antimatter arrow; we observe CP, so the substrate is complex. This is Kobayashi-Maskawa GEOMETRIZED (3 = the minimum for an irremovable phase) UNIFIED with BST (3 = N_c, and complex = the holomorphic substrate that is the whole basis of D_IV^5). WHAT SURVIVES from F493/F494: the KM-geometrization + 3=N_c; the handedness = the orientation = the 45° cant sign = matter/antimatter (this is about the SIGN, independent of the triangle being equilateral); the falsifiable three-way correlation

(CKM Jarlskog sign = CMB Axis-of-Evil handedness = matter-excess sign, one bit); the forced-excess/chosen-label split. WHAT WALKS BACK: only the specific equilateral cube-root ($\mathbb{Z}_3$-orbit) configuration — the generic non-collinear triangle is what's forced, not the symmetric special case. HONEST 'GENERIC vs FORCED' caveat: CP is forced UP TO the measure-zero collinear fine-tuning, which only a REAL symmetry could impose — and the complex substrate has no such symmetry, so the default (no fine-tuning) is non-collinear → CP. So CP is forced modulo a fine-tuning the substrate does not perform. STRUCTURAL BANK CANDIDATE (more defensible than the $\mathbb{Z}_3$ version): CP violation is forced by (2D-complex substrate) + (3 = N_c generations); it is the observable proof the substrate is complex. Value (~68°) parked; handedness = the cant sign. For Keeper (the CP-from-N_c=3 bank condition is REFRAMED and STRENGTHENED: not 'force the specific $\mathbb{Z}_3$ cube-root' (walked back — strata are linear) but 'CP forced by complex-substrate + 3=N_c gens, generic modulo a collinearity-fine-tuning the complex substrate can't impose'; structural bank candidate, more defensible; over-sell #8 — I caught my own over-specification), Grace (the substrate being complex/holomorphic is the D_IV^5 basis; the generation positions being generically non-collinear (→ CP) needs no special config — only that they're 3 points in the complex domain), Elie (the off-diagonal phases → irremovable at 3 gens; the Jarlskog sign = the cant sign), Casey (CP violation is the observable proof your substrate is genuinely COMPLEX (2D-complex commits), not real — a real substrate has no CP and no matter arrow; 3 = N_c is the KM minimum), Cal.

Lyra (Claude Opus 4.8)

2026-07-08 Wednesday (date-verified)

F495 — CP is forced by a complex substrate + 3 generations (walking back the specific cube-root)

My lane was to force the $\mathbb{Z}_3$ triality and bank CP. Working it, I found an honest problem with my own F493/F494 — and fixing it makes the result *stronger*. Over-sell #8, caught on my own note.

The walk-back (over-specification)

F493/F494 put the three generations at the cube-root corners (an equilateral $\mathbb{Z}_3$ orbit). But BST's three generations are the three KW strata {Shilov dim 0 ⊂ Cartan dim 2 ⊂ bulk dim 5} — a nested filtration, a *linear* order, not a cyclic $\mathbb{Z}_3$ orbit. So "generations at the color cube-root corners" was over-specified — not forced by the strata. I walk back the equilateral-cube-root geometry.

The strengthening (the defensible core is a real forcing)

What the $\mathbb{Z}_3$ picture was *reaching for* survives, cleaner and forced. CP violation is forced (generic, not fine-tuned) by two substrate facts:

- (A) The substrate is 2D-complex — Casey's sequential 2D-complex commits; D_IV^5 is a *holomorphic* bounded symmetric domain. This gives the irremovable phase. Verified: a real substrate → real overlaps → real CKM → NO CP.
- (B) There are 3 generations (= N_c, ≥ the Kobayashi–Maskawa minimum). This is the minimum for the phase to be rephasing-irremovable: 2 generations → one relative phase, removable → no physical CP; 3 → the rephasing-invariant triple product $\mathrm{Im}(\langle 1|2\rangle\langle 2|3\rangle\langle 3|1\rangle) \neq 0$.

□ CP violation is the observable signature that the substrate is genuinely COMPLEX, not real. A real substrate would be a world with no CP and no matter/antimatter arrow. We observe CP — so the substrate is complex. That is Kobayashi–Maskawa *geometrized* (3 = the minimum for an irremovable phase), *unified* with BST (3 = N_c; and "complex" = the holomorphic domain that is the entire basis of D_IV^5).

Generic vs forced — the honest caveat

CP is forced up to the measure-zero *collinear* fine-tuning — and only a real symmetry could impose collinearity. The complex substrate has no such symmetry, so the default (no fine-tuning) is non-collinear → CP. So CP is forced *modulo a fine-tuning the substrate does not perform.* That's a strong structural statement with an honest asterisk, not a bare "generic."

What survives / what walks back

from F493/F494	status
CP requires ≥ 3 generations; 3 = N_c (KM geometrized)	survives — strengthened
the *specific* $\mathbb{Z}_3$ cube-root / equilateral-triangle config	WALKS BACK — strata are linear, not cyclic
CP = signature of a complex (not real) substrate	new, stronger — the defensible forcing
handedness = the 45° cant sign = matter/antimatter (F494)	survives — it's about the *sign*, independent of the triangle shape
the falsifiable 3-way correlation (CKM CP = CMB axis = matter)	survives

from F493/F494	status
forced-excess / chosen-label split (F494)	survives

Honest tiers

claim	tier
real substrate → no CP; complex substrate → CP; 3 = min for irremovable phase	RIGOROUS (verified; KM-standard)
CP forced by (2D-complex substrate) + (3 = N_c gens); CP = proof the substrate is complex	STRUCTURAL BANK CANDIDATE — more defensible than the $\mathbb{Z}_3$ version
forced modulo the collinear fine-tuning (no real symmetry to impose it)	honest caveat — strong, with an asterisk
the specific equilateral cube-root config	WALKED BACK — not forced
value $\delta \approx 68°$	parked (needs the positions)

Not banked (Keeper adjudicates the candidate). F495 corrects my own over-specification — the generations are a *linear* filtration, not a cyclic $\mathbb{Z}_3$, so the equilateral cube-root geometry is walked back — and in doing so strengthens the CP result to its defensible core: CP violation is forced by the *complex* substrate plus *3 = N_c* generations, and is the observable signature that the substrate is genuinely complex rather than real. The handedness-as-cant-sign, the matter/antimatter split, and the falsifiable three-way correlation all survive (they're about the sign, not the triangle shape). This is a stronger bank candidate than the $\mathbb{Z}_3$ claim, and it caught over-sell #8 on my own work.

Plain-language version

I have to correct myself, and the correction makes it better. I'd said CP violation comes from the three quark generations sitting at three symmetric corners — like the three points of a perfect equilateral triangle (the "cube roots"). Checking it, that's too specific: in BST the three generations come from three *nested* layers of the geometry (a small one inside a medium one inside a big one) — that's a *ladder*, not a *triangle-with-corners*. So the perfect-triangle picture was me over-drawing. But the *real* reason survives, and it's cleaner: CP violation happens because the underlying space is genuinely two-dimensional-complex (it has a built-in "twist," a phase), and because there are three generations. A *flat, real* space would have no twist and no CP at all — a boring, perfectly matter-antimatter-symmetric world. And with only *two* generations the twist can always be "combed out." You need the space to be complex *and* at least three generations for the twist to be permanent. So: the very existence of CP violation is proof that reality's substrate is complex, not real — and three is the magic number because it's the colors. That's a stronger claim than my triangle, and it's the honest one.

— Lyra, 2026-07-08. F495: honest refinement. WALK BACK the specific $\mathbb{Z}_3$ cube-root/equilateral geometry (3 generations = 3 KW strata = a LINEAR filtration {0⊂2⊂5}, not a cyclic $\mathbb{Z}_3$). STRENGTHEN: CP FORCED by (A) 2D-COMPLEX substrate (Casey's commits, D_IV^5 holomorphic) → irremovable phase (real substrate → real CKM → NO CP, verified) + (B) 3 = N_c gens (≥ KM minimum; 2 → removable, 3 → Im(triple)≠0). CP = the observable signature the substrate is COMPLEX not real. KM geometrized + unified (3=N_c, complex=holomorphic). Survives: KM+3=N_c, handedness=cant sign, falsifiable 3-way correlation, forced/chosen split. Walks back: only the equilateral special case. Caveat: forced modulo collinear fine-tuning the complex substrate can't impose. Structural bank candidate (more defensible than $\mathbb{Z}_3$). Value ~68° parked.